

A definite pencil beyond quotients of individual spectra

Counterexample to Conjecture 00000007177

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Precise assertion and matrices

The source states that generalized eigenvalues have the scale of the quotient of the two spectra. We refute the individual-spectral quotient assertion: a pencil eigenvalue need not equal α/β for any individual eigenvalues α of A and $\beta \neq 0$ of B . Consider

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Both matrices are real symmetric positive definite. Indeed, for a nonzero $z = (x, y)^T$,

$$z^T A z = x^2 + 2y^2 > 0, \quad z^T B z = 2x^2 + 2xy + 3y^2 = x^2 + (x + y)^2 + 2y^2 > 0.$$

Thus the example lies within the definite-pair setting.

An actual pencil eigenvalue

The standard generalized eigenvalue condition for the pencil $A - \lambda B$ is a nonzero vector with $Av = \lambda Bv$. Here

$$Av = \begin{pmatrix} 1 \\ -2 \end{pmatrix} = Bv,$$

so $\lambda = 1$ is a genuine generalized eigenvalue. Neither the vector equation nor definiteness is assumed.

No quotient of the individual spectra

For $Az = \alpha z$ with nonzero $z = (x, y)^T$, the two coordinate equations are $(\alpha - 1)x = 0$ and $(\alpha - 2)y = 0$. At least one coordinate is nonzero, so $\alpha \in \{1, 2\}$. Conversely, the two coordinate vectors realize 1 and 2, proving that the complete individual spectrum of A is $\{1, 2\}$.

Neither number is an eigenvalue of B . If $Bz = z$, its equations give $x + y = 0$ and $x + 2y = 0$, forcing $x = y = 0$. If $Bz = 2z$, its equations give $y = 0$ and $x + y = 0$, again forcing zero. Thus neither equation has a nonzero eigenvector. Now if $1 = \alpha/\beta$ with $\beta \neq 0$, then $\alpha = \beta$. An individual spectral quotient equal to 1 would therefore require A and B to share an eigenvalue, which the preceding arguments exclude. The generalized eigenvalue 1 is absent from every possible quotient of their individual spectra, irrespective of how those spectra are ordered.

Distinction from the Rayleigh quotient

The generalized Rayleigh quotient uses the same vector in both forms:

$$R(z) = \frac{z^T A z}{z^T B z}.$$

It is not a quotient of two individual eigenvalues. In this example $v^T A v = v^T B v = 3$, so $R(v) = 1$, exactly as expected. The counterexample does not dispute this valid variational formulation or spectral interval bounds. It disproves the source's asserted individual-spectra quotient description; the separate extremum clause is unnecessary for refuting the conjunction.

Formal verification

Lean defines actual matrices, matrix-vector multiplication, eigenvalues through nonzero eigenvectors, and the pencil condition $Az = \lambda Bz$. It proves symmetry and positive definiteness for all nonzero vectors, the concrete pencil witness, the full individual spectrum of A , the two exclusions for B , and impossibility of every quotient representing 1. The final theorem packages these properties without assumed spectral tables or supplied definiteness data.